

23CS504	CLOUD INFRASTRUCTURE AND SERVICES MANAGEMENT	L	T	P	C
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COURSE OBJECTIVES

1. To develop practical knowledge of cloud infrastructure and virtualization technologies.
2. To enable deployment and management of cloud services using industry platforms.
3. To build skills in containerization and microservices using Docker.
4. To apply DevOps practices for continuous integration and deployment.
5. To analyze and implement scalable and secure cloud-based solutions.

LIST OF EXPERIMENTS

1. Virtualization Setup
 - Install and configure VirtualBox/VMware
 - Create and manage virtual machines with different OS
2. Virtual Networking
 - Configure NAT, Bridged, and Host-only networking
 - Enable communication between virtual machines
3. AWS Environment Setup
 - Create AWS account
 - Explore AWS Management Console and services
4. Compute Services
 - Launch and configure EC2 instances
 - Connect using SSH and manage instances
5. Cloud Storage
 - Create and manage S3 buckets
 - Upload, download, and configure permissions
6. Identity and Security
 - Configure IAM users, roles, and policies
 - Implement basic cloud security practices
7. Docker Installation and Basics
 - Install Docker Engine
 - Create, run, stop, and remove containers
8. Docker Image and Networking
 - Create Docker images using Dockerfile
 - Configure Docker bridge networking
9. Multi-Container Deployment
 - Use Docker Compose for multi-container applications
 - Deploy a simple web application
10. DevOps and Deployment
 - Implement basic CI/CD workflow
 - Deploy containerized application on cloud/local system

TOTAL: 60 PERIODS

COURSE OUTCOMES

Upon completion of the course, students shall have ability to

CO1	Configure and manage virtualized environments and cloud infrastructure.	[AP]
CO2	Deploy and manage compute and storage services in cloud platforms	[AP]
CO3	Develop and deploy containerized applications using Docker	[AP]
CO4	Implement basic DevOps practices for automation and deployment.	[A]
CO5	Design scalable and secure cloud-based solutions for real-world applications.	[C]

TEXT BOOKS

1. Michael J. Kavis, Architecting the Cloud: Design Decisions for Cloud Computing Service Models, Wiley, 2014.
2. Nigel Poulton, The Docker Deep Dive, 2023 Edition.
3. John Arundel & Justin Domingus, Cloud Native DevOps with Kubernetes, O'Reilly, 2019.

REFERENCE BOOKS

1. Rajkumar Buyya, *Mastering Cloud Computing*, McGraw Hill, 2013.

WEB RESOURCES

1. AWS Official Documentation – <https://docs.aws.amazon.com>
2. Docker Documentation – <https://docs.docker.com>
3. Kubernetes Learning Portal – <https://kubernetes.io/docs/tutorials/>
4. Cloud Computing Tutorials – https://www.tutorialspoint.com/cloud_computing